Design Brief Rubric

Assignment: Design Brief for Engineering Project
Due Date: As assigned (via Google Classroom)
Total Points: 100
Purpose: Evaluate the team's ability to apply the Ask, Research, and Imagine steps of the EDP, produce a clear design brief, and use the Pugh Matrix to select a solution for an engineering project. Individual contributions are assessed via journal entries.

Criteria	Exceeds Expectations (90–100%)	Meets Expectations (80–89%)	Approaches Expectations (70–79%)	Needs Improvement (<70%)	Points (25 max per criterion)
Problem Statement (Ask Step)	Clearly defines the project problem, identifying specific stakeholders, needs, and constraints (e.g., budget, materials, time). Thoroughly integrates ethical considerations (e.g., sustainability, safety). Questions for the "customer" (instructor or stakeholder) are insightful and directly tied to project goals.	Defines the project problem with stakeholders, needs, and constraints. Includes ethical considerations. Questions for the "customer" are relevant but may lack depth or specificity.	Problem statement is vague or misses some elements (e.g., stakeholders, constraints). Ethical considerations are minimal or generic. Questions for the "customer" are incomplete or lack focus.	Problem statement is unclear, missing key elements, or lacks ethical considerations. Questions for the "customer" are absent or irrelevant.	/25

Research Summary (Research Step)	Summarizes 3–5 credible sources (e.g., technical guides, industry articles) with detailed insights relevant to the project's goals and constraints. Clearly connects research to project needs and ethical considerations (e.g., environmental impact). Citations are complete, including AI tool use if applicable (per Lab Rules Section 9).	Summarizes 3–5 sources with relevant insights into project solutions or components. Connects to constraints and ethics. Citations are mostly complete but may miss minor details (e.g., AI tool date).	Summarizes fewer than 3 sources or lacks depth in insights. Connections to constraints or ethics are weak or unclear. Citations are incomplete or missing AI use details.	Research is minimal, irrelevant, or lacks credible sources. No clear connection to project needs or ethics. Citations are absent or incorrect.	/25
Brainstormed Ideas (Imagine Step)	Presents 5+ creative, diverse ideas with detailed descriptions and clear, labeled paper/pencil sketches (photos uploaded or described). Ideas fully address project constraints and ethical considerations. Team collaboration is evident through detailed journal entries documenting individual roles.	Presents 5 ideas with clear descriptions and adequate sketches. Ideas address constraints and ethics. Collaboration is evident in journals but may lack detail on individual contributions.	Presents fewer than 5 ideas or ideas lack creativity/detail. Sketches are unclear, incomplete, or missing. Constraints or ethics are minimally addressed. Collaboration is unclear in journals.	Ideas are minimal, vague, or lack sketches. Constraints or ethics not addressed. Little to no evidence of team collaboration in journals.	/25

Evaluation (Pugh Matrix)	Pugh Matrix compares all ideas to a baseline using 3–5 weighted criteria (e.g., functionality, cost, feasibility, ethics). Math is accurate (scores: +1, 0, -1 × weights, summed correctly). Justification for the top solution is clear, logical, and tied to project goals and ethical considerations.	Pugh Matrix compares ideas to a baseline with 3–5 criteria. Math is mostly accurate (minor errors). Justification for the top solution is provided but may lack full clarity or strong ethical focus.	Pugh Matrix is incomplete (e.g., <3 criteria) or contains math errors. Justification for the top solution is weak, vague, or lacks ethical tie-in.	Pugh Matrix is missing, incorrect, or lacks sufficient criteria. Math errors are significant. No clear justification for the selected solution.	/25

Additional Notes

- **Total Score**: Sum of points across criteria (out of 100). Contributes to Projects (40%) in syllabus grading.
- **Versatility**: This rubric applies to any design brief for engineering projects (e.g., robot designs, mechanisms, or EDP-based tasks). Criteria and constraints should be specified in the project prompt (e.g., budget, materials, or timeline).
- **Teamwork**: Assessed through the group-produced brief and individual journal entries (Participation & Journals, 30%). Journals must document each student's role (e.g., problem definition, research, sketching, Pugh Matrix calculations).
- **Ethics**: Ethical considerations (e.g., sustainability, safety, societal impact) must be explicit in Problem Statement and Evaluation sections (Syllabus Objective 12; Excel I.A.3).
- **Citations**: Per Lab Rules Section 9, cite all sources, including AI tools (e.g., "Generated with Grok by xAI, [date]"). Include citations in the Research Summary section.
- **Format**: Submit as a Google Doc with clear sections (Problem Statement, Research Summary, Brainstormed Ideas, Evaluation), uploaded sketches (photos or descriptions), and a Pugh Matrix table. Follow the project-specific template provided.
- **Feedback**: Scores and comments will be provided via Google Classroom. Late submissions: -10% per day (max 3 days, per syllabus).
- **Adaptation**: For projects with unique requirements (e.g., specific criteria for Pugh Matrix), instructors may adjust criteria weights or add project-specific expectations while maintaining the 100-point scale.